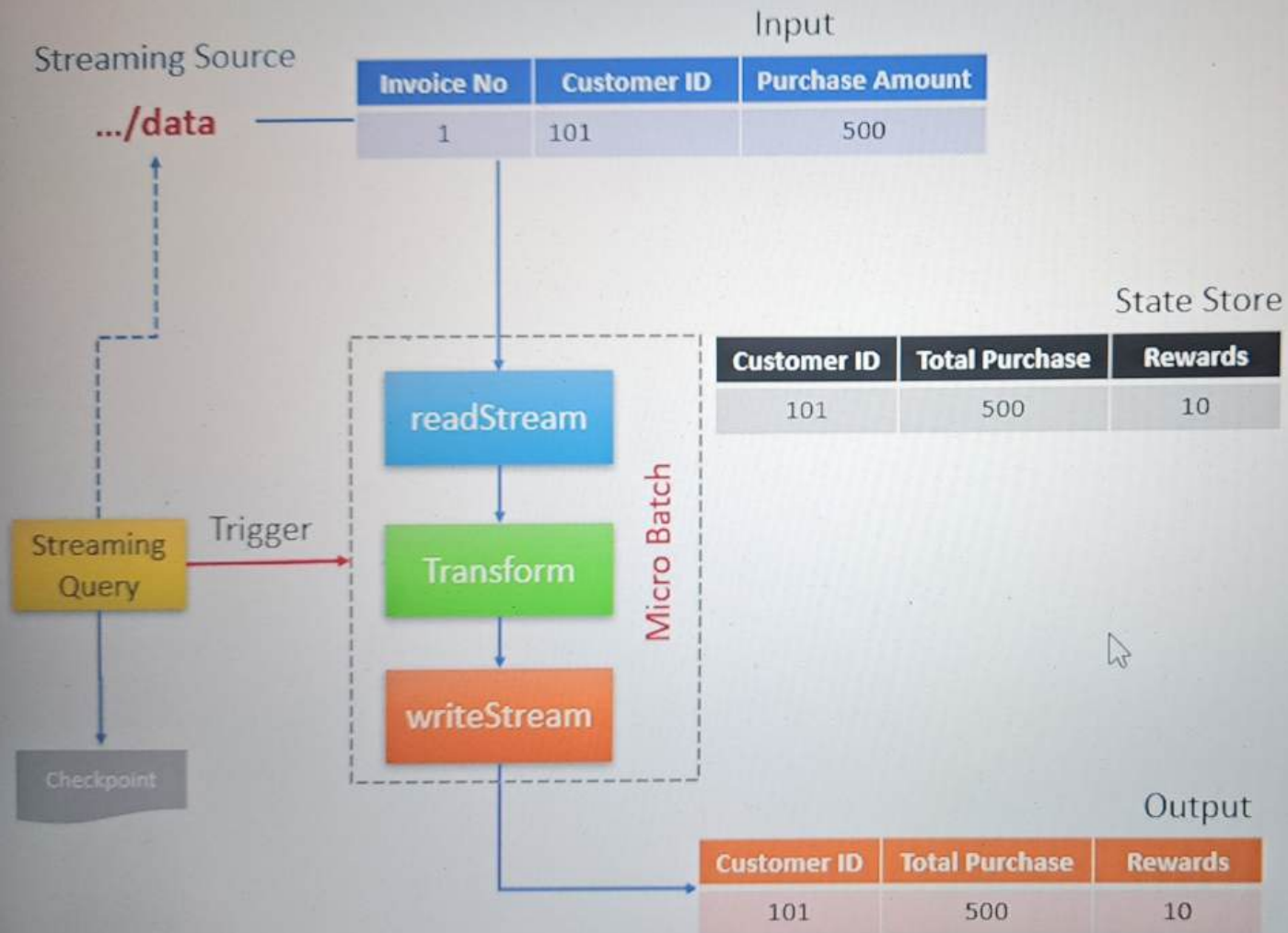
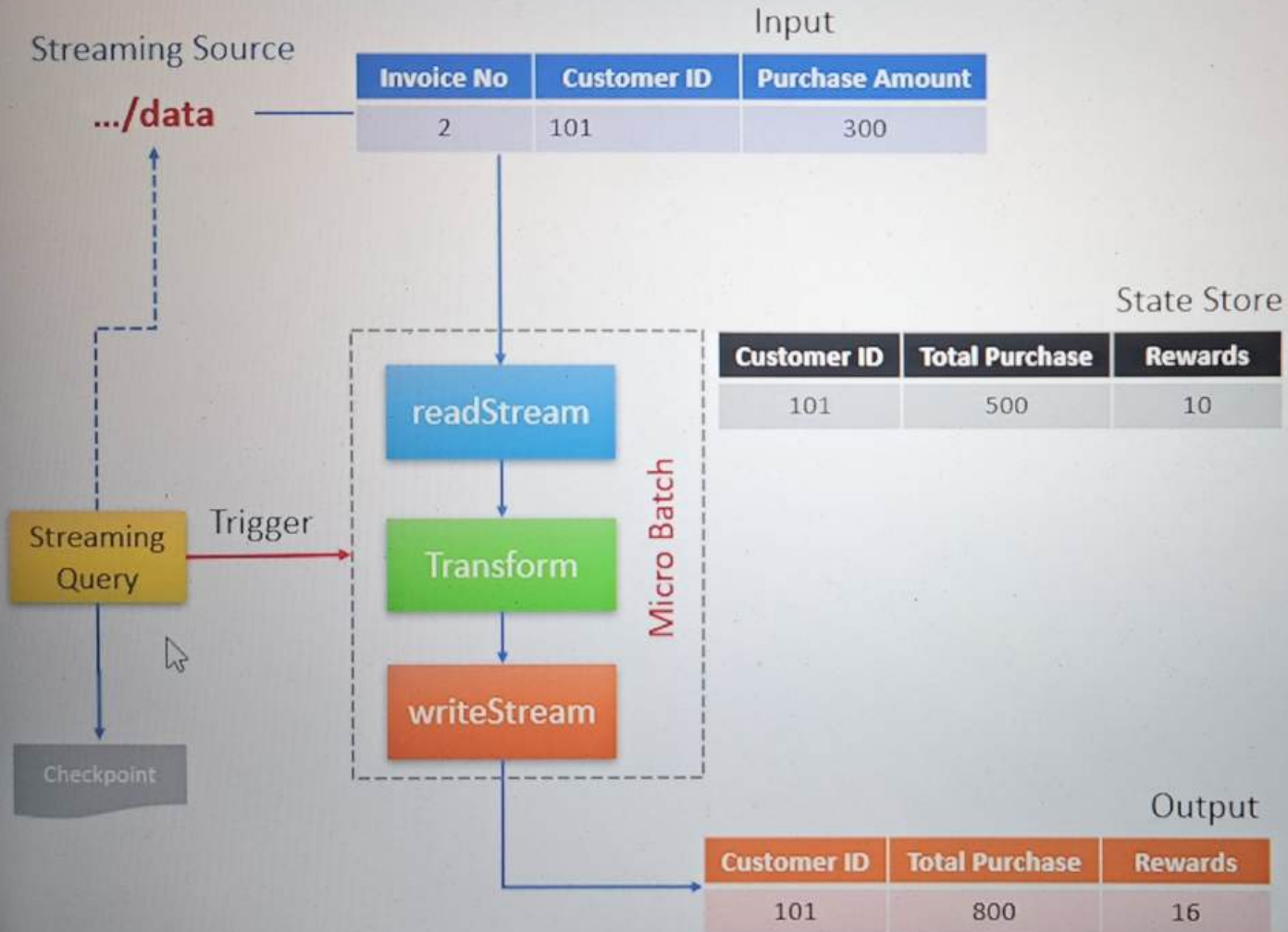


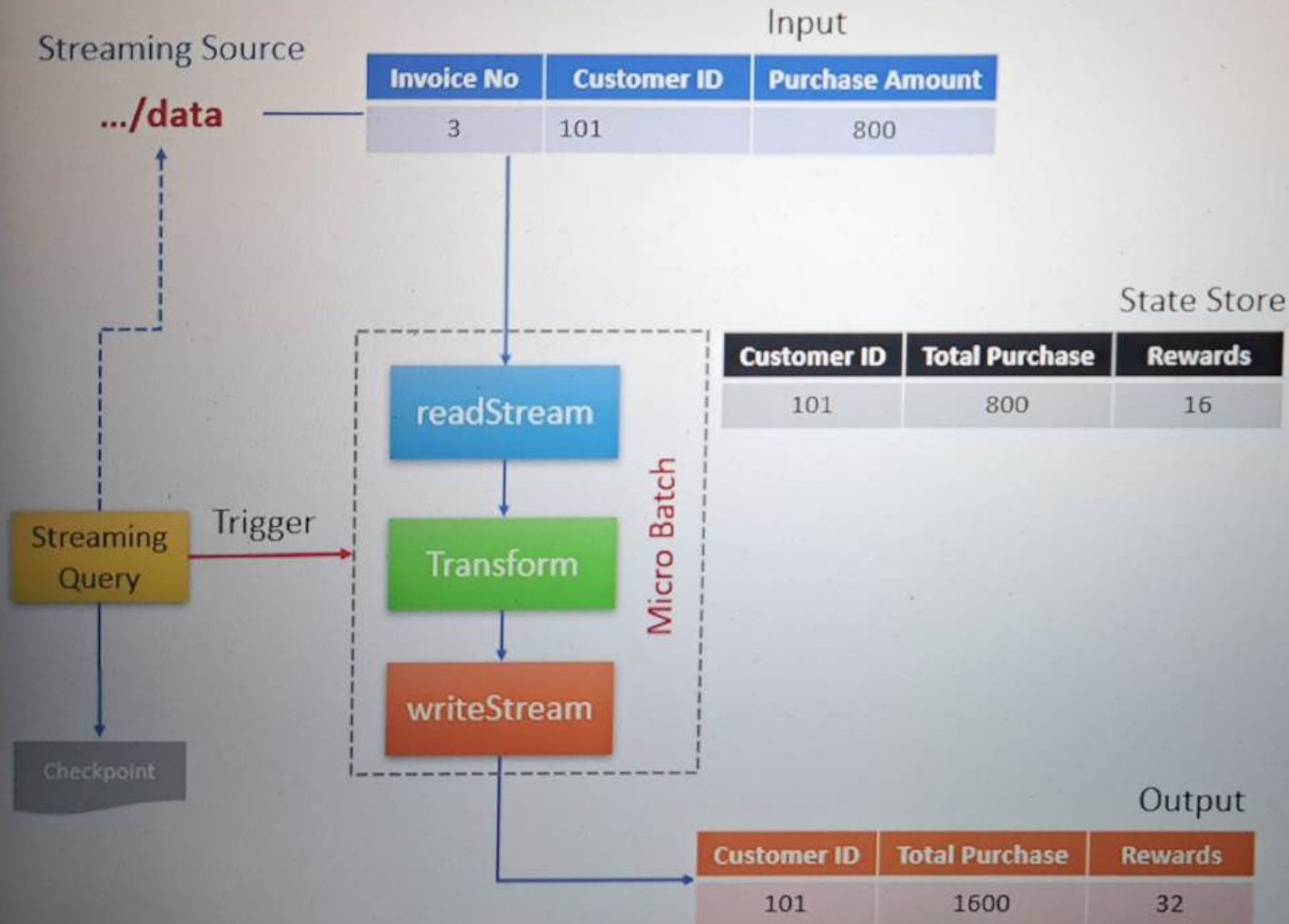
What is the state?



What is the state?

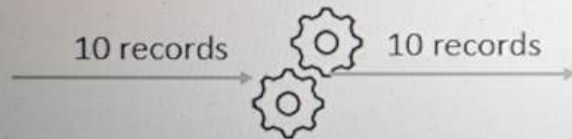


What is the state?



Stateless Transformation

- `select()`, `filter()`, `map()`, `flatMap()`, `explode()`, etc.
- It does not support Complete output mode



Stateful Transformations

- grouping, aggregation, windowing, joins
- Excessive state may cause OOM

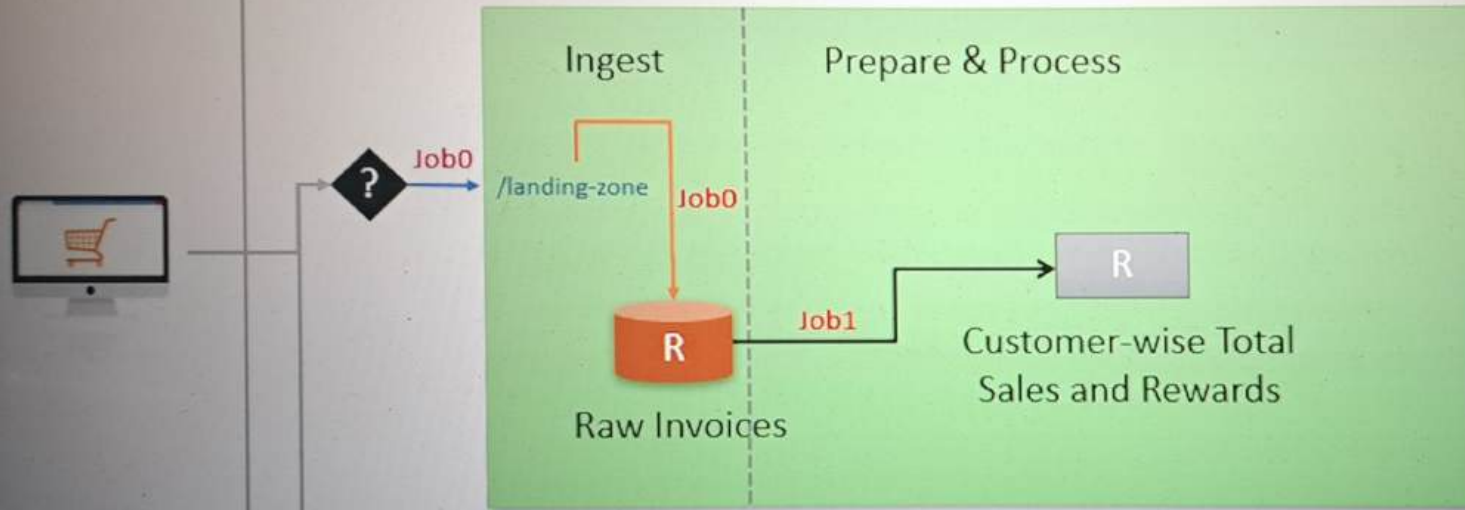
- Managed Stateful Aggregations
- Custom Stateless Aggregations



Source System

Data Engineering Platform

Consumer Application



Unbounded Continuous Aggregation

Compute Loyalty Points for the customer

Timebound Aggregation

Compute Monthly Loyalty Points for the customer

